



Pittala Sree Sai Pavan

Roll No.:240102070

B.Tech - Electronics and Communication Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology Guwahati	6.86 (Current)	2024-Present
Senior Secondary	Telangana Board of Intermediate Education	96.2%	2024
Secondary	Telangana Board of Secondary Education	97%	2022

PROJECTS

- **Two-Stage Miller-Compensated Operational Transconductance Amplifier** April 2026
Analog Integrated Circuit Design LTspice
 - Designed a **two-stage OTA** in **TSMC 0.18 μm CMOS** achieving **71 dB DC gain** and **GBW $\approx$ 14 MHz** at 1.8 V supply.
 - Applied **Miller compensation** with **zero-cancellation**, yielding **88° phase margin** — 28° above the stability criterion.
 - Sized six **MOSFETs** via **gm/ID methodology** ($V_{ov} = 200\text{ mV}$); verified **closed-loop gain of 2 $\times$** through **step response**.
 - Implemented a **10 $\times$ NMOS current mirror** for tail biasing, achieving **100 μA tail current** at **0.38 mW** total power.
- **Traffic Signal Controller on FPGA** March 2026
Digital Logic Design Verilog HDL
 - Designed a **2-state synchronous FSM** in **Verilog HDL** for a 2-path traffic controller with **5-second** phase sequencing.
 - Implemented a **clock divider** module scaling the system clock down to **1 Hz** for accurate synchronous state transitions.
 - Synthesised and deployed onto a **Zedboard (Zynq-7000) FPGA** via **Vivado**; validated **LED sequencing** on hardware.
- **Automated Circuit Analyser** February 2026
Computational Electrical Engineering Python, MATLAB
 - Built a **Python** circuit solver using **NetworkX** to model circuits as derected graphs for systematic **KCL/KVL** analysis.
 - Generated **Cut Set** and **Tie Set** matrices automatically from **spanning tree** to formulate independent circuit equations.
 - Solved **Laplace-domain** impedance matrices via **SymPy**, computing s-domain voltages and currents across all branches.
 - Applied **inverse Laplace transforms** to obtain **time-domain transient responses**, visualised using **Matplotlib** for all nodes.

TECHNICAL SKILLS

- **Hardware & Simulation:** LTspice, Verilog HDL, MATLAB, Simulink*
- **Programming Languages:** C, C++, Python, HTML, CSS, JavaScript
- **Tools/Libraries:** NumPy, Pandas, Matplotlib, sympy, Networkx, Git, GitHub
- **Operating Systems:** Windows, Linux

* Elementary proficiency

RELEVANT COURSEWORK

- **Core Electronics:** Basic Electronics, Digital Circuits (Theory & Lab), Circuit Theory, Signals and Systems, Analog Circuits, Control Systems
- **Mathematics:** Linear Algebra, Calculus, Differential Equations, Real Analysis, Complex Analysis, Probability and Random Processes

ACHIEVEMENTS

- **Competitive Programming**, Pupil on Codeforces with a max-rating of **1350**; **CF: ssp_09** 2025–Present
- **JEE Advanced 2024**, Ranked in the **top 2.53%** among 1.8 lakh+ candidates across India 2024
- **TS EAPCET 2024**, Ranked in the **top 0.87%** among 2.4 lakh+ candidates across India 2024

EXTRACURRICULAR ACTIVITIES

- **CoS, NSS Volunteer**, NSS Cell, IIT Guwahati: Contributed 30+ hours to community service activities, including village visits, face painting, and engagement programs. Jan. 2026 - present
- **Junior Web Developer**, Students Web Committee — frontend dev on institute projects Oct. 2025 - Present
- **PR & Branding Associate**, Alcheringa — Northeast India's largest cultural fest Nov. 2024 - Feb. 2025